

Institute of Actuaries of India

Subject CP1– Actuarial Practice (Paper B)

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Assignment 1**Solution 1:**

Factors

- It needs to be assessed whether the market share will really improve on having online sales. or [0.5]
- Whether sales would move from agency channel/broker channel to this new channel, hence no new incremental sales happen [0.5]

Is there any other reason for other companies to have better market share? [0.5]

- Any special features in their products [0.5]
- for eg., add on benefits like Nil depreciation cover, tyre cover etc [0.5]
- Does ABC Ltd have price disadvantages due to high overheads, high commission to intermediaries [0.5]
- Is other company selling motor product as a loss leader [0.5]
- Do other companies offer better value to customers through better services, better claims practices/interface etc. [0.5]

Cost versus benefit analysis [0.5]

Cost versus benefit analysis has to be done whether there will be enough sales through this product which will compensate for the costs involved in designing, pricing, developing and launching this product. [1]

Costs of setting up an online platform/other IT systems to cater to the needs of this segment which will involve significant capital investments [1]

As all policy holder's data need to be handled online, any existing guidelines and regulations regarding data, security issues need to be adhered to, which would involve one time/recurring costs. [1]

As there is no past experience in this type of business, ABC Ltd may have to hire consultants, take reinsurance and/or outsource the IT aspects which will again involve costs. [1]

Decision needs to be taken on what type of customers should be targeted, what type of motor vehicles (for eg., <3 year old cars or older cars) ? [0.5]

Covering old vehicles could be a problem as some level of assessment of state of vehicle at time of insurance needs to be considered. Hence this could lead to additional costs. [0.5]

On line questionnaire has to be designed to capture standard details of vehicles, call has to be taken as to the level of details that would be sought [0.5]

Compromise has to be done between making the entire on line purchase a simple process for customer to ensuring required risk/rating factors are captured for pricing [0.5]

Auto underwriting/quote system needs to be designed suitable for this channel. Some level of integration could be required with centralised data bases like – [0.5]

- Vehicle registration bureau (to make sale process easier)
- Credit bureau
- Industry data base to automatically determine accident records, NCB rate, fraud claims etc

[any one example 0.5 Marks]

Channel conflicts need to be managed, as tied agency may feel that the channel is encroaching upon their sales [0.5]

If price of online product is very attractive it could lead to lapse of regular motor policy purchased through other channel which would also lead to channel conflict [0.5]

In the medium to long term, if this channel is successful, it could lead to redundancies of the existing employees, which would be of concern to management [0.5]

Based on the costs and other considerations, the company has to assess whether they have the sufficient capital to launch this product. [0.5]

Are these the best use of company's resources including human capital? [0.5]
[8 Max]

Solution 2:

Pricing Assumptions:

They may not be able to sell the same product with same price through internet and may need to re-price the product due to the following reasons:

Target market [4]

It could be a different target market. [0.5]

On line channel more attractive to Younger age than an older age group [0.5]

who are likely to have a different risk profile [0.5]

Frequency likely to be different

- more urban, younger population etc. [0.5]
- Frequency could also be higher due to –

[0.5 mark per point]

1. Relatively Lower income profile of young customers - may even claim for small amounts
2. May not bother much about NCD etc if they are more concerned about immediate financial relief
3. Claims process could also be simpler for on line process compared to existing process which may encourage policyholder to register and follow up on a claim
4. are also likely to be more active users of vehicles and hence expect more than average mileage/hence greater exposure to risk

Severity could be higher due to-

[0.5 mark per point]

1. More reckless in driving resulting in more claims from total loss
2. Socio-economic profile could be different leading to higher claims from theft
3. Risk of frauds higher as the class of customers are likely to be less economically stable than older customers

[1 Max]

Expenses:

The product will not have expense relating to commission as with existing product.

[0.5]

Expenses will be different – New IT systems/online platforms are to be developed, maintained, more publicity costs etc.

[1]

Policy illustrations, proposal, policy document etc. should be made to cater for online purchases.

[0.5]

The expense per policy loading will depend on the expected business volumes.

[0.5]

There has to be trade-off between loading the real expenses and keeping the product competitive.

[0.5]

[2 Max]

Renewal:

Since it is a product initiated by the customer it could have a better persistency.

[0.5]

Although competition could make it less so

[0.5]

Renewal may be lower than other channel as customers being tech savvy will shop around every year for a better quote

[0.5]

Renewal could be also be poor if company does not have good systems/process to follow up with clients and get policy renewed in the absence of intermediary

[0.5]

As there is no intermediary to explain this product, customers may not understand the product- coverage/exclusions

[0.5]

Likely higher chance of claims being denied/settle for lower amount than expected. Customer may withdraw if they feel that the product does not suit their needs.

[0.5]

[2 Max]

[8 Max]

Solution 3: At times of economic recession, policyholders are more concerned about immediate relief rather than any medium/long term benefits of not claiming like a NCD

[0.5]

Increase in claim frequency:

[1]

- Customers may claim even for small amounts
- Theft rates could go up
- Deliberate acts of damage to vehicle to get monetary benefit

Increase in claim severity:

[1]

- Claim amount would be inflated by customers
- Theft claims leading to increase in total loss claims

Increased crime or fraudulent claims:

[2]

- Customers may collude with the motor service centres and fake claims or inflate claims.
- Staged claims - Third party property damage/disability claims

[0.5 mark each]

[4 Max]

Solution 4: Claims Underwriting

Underwriting at the claim stage can help to both reduce excessive claims and identify fraudulent claims.

However the costs of implementing and maintaining a control system must be compared with the benefits gained from it.

The level of scrutiny of claims should be linked to claim amount and the claim type.

For eg., a liability claim should undergo more scrutiny than an own damage claim or a Total Loss/theft claim should be scrutinized more than a damage claim

Claims cost management

For management of overstatement of claims costs

- the company can get the estimates vetted by the experts in their panel or
- Have a readymade list of likely cost ranges for each type of repair/parts that need replacement for all model types or
- insist on getting the motor vehicles repaired at the garages/service centres approved by them.

At a further level the insurer might require that damage is inspected by one of its employees or agents before remedial work is authorised.

Penalties for frauds

Penalize any sales advisor whose clients are consistently claiming (due to fraud)

Industry level co-operation to manage frauds

Data sharing:

- Data is shared with other insurers regarding individuals who have made fraudulent claims in the past.
- Liaison with reinsurers and authorities to catch fraudulent customers
- To be a member of any fraud tracking agency and get the list of perennial fraudulent customers or garages/service centres with whom these fraudulent claims are deducted.
- Offer rewards to garages or brokers/agents for deducting fraud

[0.5 mark each]

[5 Max]

Solution 5:

Asset values:

[1]

- Fall in asset values due to increased risk of default and hence more provisions for non-performing assets.
- Market value of equities/property could fall and hence reducing the asset values, if market valuation is being considered for solvency demonstration.

Government Bonds:

[1]

- Government borrowing would increase during recession to meet the government expenditure.
- It could lead to a larger supply of government fixed-interest bonds, increase in interest rates and so to reductions in the prices of fixed-interest bonds.

Corporate Bonds:

[1]

- The yield margin of corporate over government bonds primarily reflects differences in marketability and security –default risk.
- Investors' perceptions of default risk may increase in times of recession as
 - Corporate profitability is reduced.
 - it is more likely that some borrowers may be unable to meet their obligations.
- yield margin increase as the economy moves into recession.

Equity:

[1]

- Domestic stock market is likely to perform badly.
- Share prices are unlikely to increase if companies are struggling to maintain profitability.
- The risk involved in holding equities may increase during a recession because:
 - the chances of company failure are higher
 - cashflow problems are more prevalent
 - default on dividends is more likely
 - dividend income is likely to be more volatile.

Property:

[1]

- Property prices may witness decreasing trend due to affordability/borrowing capacity of the buyers.
- Market sentiment may not be conducive to invest in properties with alternate attractive investment options diverting the customers away from property market
- There could be an expectation of fall in property prices leading to poor investment sentiment.

Rental income:

- Default in rental payment by tenants.
- Business activities in general may get reduced resulting in lower demand for property on rent for both commercial and residential purposes.

Money market/cash investments:

[1]

- Cash would give a reasonable income stream combined with no risk of capital loss, unlike equities and fixed-interest investments, which may impact the principal invested.

[6 Max]

Solution 6: Policy Liabilities

Higher Claims costs:

[1]

Expected Higher claims by frequency and severity

Further increase in provisions for future claims due to the above.

Credit for Reinsurance in provisions:

[1]

Default by reinsurer due to recession and not able to get a suitable new reinsurance arrangement. This would lead to increase in net provisions as

recoverability from re-insurers could be suspect

Regulator:

Regulator might also ask for increased the margins on provisions which could also lead t increase in provisions [0.5]

Expenses

Lower new business/renewals as the policyholders are not able to afford the premiums. [0.5]

Expenses could increase due to:

- Higher expenses due to monitoring/controlling fraud claims
- Higher expenses due to higher number claims
- Due to likely lower business overheads are to be spread over lesser number of policies.

[0.5 mark per point] [1 Max]

All this could lead to higher provision for expenses in IBNR/IBNER and URR calculations. [0.5]

[4 Max]

Solution 7: Outline an investment strategy most suitable for this scheme [6]

Investment strategy:

The investments should be appropriate to the nature, term, currency and uncertainty of the liabilities and should take into account the provider's appetite for risk. [0.5]

Subject to the above, the investments should be selected to maximise the overall return (income plus capital) on the assets. [0.5]

Currency of liability:

As all contributions, expenses and payments will be in domestic currency, the investments will also be in domestic currency. [0.5]

Nature of Liability:

Liability is linked to projected salary at retirement. Salary inflation normally tends to be higher than price inflation [0.5]

Type of investment:

Domestic equities are broad match for active members salary linked benefits in the long term. [0.5]

Property: broad match for real liabilities and a diversification from equities. [0.5]

Rent can be used to meet expenses (provided there are other investments). [0.5]

Property investments have liquidity issues, require large investments and may be invested if the fund size is large. [0.5]

Deferred Members:

Nature :As there is no salary growth post exit, investments could be made in domestic government conventional bonds/high rated corporate bonds and nearing retirement it is switched to cash or money market investments. [1]

Term:

[1 Max]

The timings of payment for a member is known as it is on attaining age 60, but there will be some level of uncertainty associated with deaths/disability before retirement

Since the incidence of death/disability is small hence the quantum cash outflow due to these reasons is likely to be relatively small compared to retirement benefits.

Maturity of scheme

[1 Max]

Some level of liquidity/cash could be required to be held especially if benefit payouts are expected to outpace contributions.

This would happen when scheme is fairly mature and there are fewer newer members and significant outflow due to retirements is expected.

Expenses :

[1 Max]

Nature : Will be real and linked to salary growth and price inflation.

[0.5]

Expenses will typically be recurring in nature and hence index linked bonds could also be considered.

[0.5]

[8 Max]**Solution 8:**

The Surplus/deficit arises due to actual experience being different from assumed in prior valuation exercise.

Mortality:

[2 Max]

Mortality profit would have arisen if the actual number deaths in service are fewer than assumed in the valuation and the average benefit paid out in respect of death claims is lower than the average accrued liability or

[1]

Hence profit could be due to more deaths among employees with high accrual of service (as only 10 years annual salary is payable) or

due to more death/s of low earning employees (as average benefit is lower than expected)

[1]

Salary Growth:

[2 Max]

If the actual salary growth has been more than assumed in the valuation it would lead to loss.

It would have led to higher benefit for the retirements in that year.

It would have led to deficit due to higher provision for future retirements

Investment return:

[2 Max]

Lower investment return than assumed in the previous valuation would have led to the deficit.

[0.5]

There could also be one of large one off loss on sale of an investment due to

- strategic sale or
- recognition of credit defaults or
- impairment of assets.

Withdrawal:

[2]

In case of withdrawal/ exit of employees, the benefit would depend on the present salary and present service and benefit payout is deferred.

The benefit will be lower on withdrawal than for a continuing member as there will not be any link to future increase in salaries.

Hence higher than expected withdrawals would have led to surplus.

[8 Max]

Assignment 2

Solution 1: The needs fulfilled and benefits of buying the product for the policyholder are as below:

- A policyholder will effect a unit-linked policy to save or build up a lump sum for retirement
- The policyholder may not be financially dependent on anyone post retirement
- The guaranteed annuity will help to take care of self and family needs post retirement
- It will help in maintaining desired standard of living post retirement
- The return on unit linked plan may be higher than a savings banking account or fixed deposits
- The return may outpace the inflation enable him to have higher annuity or lump-sum on retirement
- The lump sum money available at retirement can help deal with emergency situations like health related costs, paying a mortgage etc.
- However the investment risk will be borne by the policyholder
- The downside risk is protected by the capital guarantee offered under the product
- Since no death benefit is available it will increase the return on the fund
- There may be tax advantages of buying the product

[0.5 mark for each]

[5 Max]

- Solution 2:**
- The guarantee is > 0% as fund management, guarantee and other charges recovered from fund/premium will also be covered under the guarantee.
 - Since cost of guarantee charge is a % of fund value, risk of lower than expected investment performance will lead to lower cost of guarantee charges
 - However the guarantee on maturity is more likely to bite when investment return is low
 - Or due to poor business mix like low ticket size and fixed administration charges levied could significantly erode premium/fund
 - There could be error in pricing of the cost of guarantee
 - or guarantee may not be priced prudently
 - Risk of selective withdrawals
 - The policyholders may not withdraw if the fund value is low close to maturity (i.e.- guarantee is in the money) leading to higher than expected cost
 - Policyholders may be more likely to lapse when their guarantees are significantly out of the money leading to the loss of future charges for the insurer
 - High new business strain due to high sales particularly if product is considered innovative in the market
 - Customers might not appreciate guarantee enough leading to Lower than expected business volume leading to difficulty in managing the funds offered with the product which could lead to poor returns
 - The guarantees in adverse scenarios can lead to higher reserving/capital requirement impacting shareholder return
 - In extreme cases and depending on the volume of business written, it could lead to insolvency of the insurer
 - For example adverse movement of the interest rates close to maturing block

of business

- The poor solvency position could lead to reputation risk for the insurer
- Change in the regulation leading to higher provisioning for the guarantee
- The high market volatility will increase the likely cost of meeting the guarantee
- Volatility in surrenders/policy holder behaviour can lead to difficulty in any hedging strategy applied to manage the risk
- Liquidity constrains if the guarantee is not adequately provisioned as shareholder will have to provide shortfall at inopportune time
- However most policyholders will not have the detailed knowledge to optimally select against the insurer and their actions will instead be driven by personal circumstances.

[0.5 mark for each] **[Max 9]**

Solution 3:

- Under economic capital the liability is sum of the expected value of unpaid liability stated on a present value best estimate basis plus a risk margin. [0.5]
- Risk Margin covers financial and non-financial risk [0.5]

Allowance for financial risk:

- Financial risk in respect of liability cash flow is normally allowed for in a market consistent calculation by replicating portfolio or [0.5]
- Stochastic modelling and suitably calibrated asset model [0.5]
- Financial risk is implicitly allowed for by taking the market value of replicating assets
- The insurance contract has embedded guarantee as the pay out on maturity could be seen as payment of underlying value of assets(A) at maturity subject to a minimum of return of premiums paid (B) [0.5]
- This can be taken as put option on A with a strike price of B [0.5]
- Unfortunately, finding a replicating portfolio is not always easy especially when dynamic policyholder behaviour and demographic risks has to be taken into account [1]
- Under the stochastic methodology- The best estimate liability required to cover the expected cost of the guarantee could likely be calculated using a stochastic approach [0.5]
- where investment return will be modelled stochastically [0.5]
- The other variables like lapses could also be projected stochastically however it will complicate the model [0.5]
- The investment return likely to be market consistent and will be modelled using probability distribution [0.5]
- Allowing appropriately for volatility and correlation [0.5]
- generated for both equity and debt component [0.5]
- A large number of simulations would be run [0.5]
- The projected cash flows should allow for all expected decrements, expenses etc. including discontinuance [0.5]
- All non-financial assumptions should be best estimate with no prudential margins [0.5]
- The liability on the best estimate basis would be the average of present value over all of the simulations [0.5]

Allowance for non-financial risk

- By holding an explicit extra provision or risk margin [0.5]
- These allowances will depend on [1]
 - The amount of risk
 - The cost of the risk implied by market risk preferences

[10 Max]**Solution 4:**

- The annuities may represent poor value due to fall in interest rate and increase in the longevity
- To provide more flexibility to the policyholder in the use of the retirement funds
- As a result of flexibility people may contribute more money in their pension funds
- It trusts the people to manage their own fund for generation of income post retirement
- It may want to enable substantial retirement savings to be directed toward other assets like property market which could also generate regular monthly income for retirees
- It may also give boost to the tax receipt particularly in the short term
- The annuity provider market may not be developed leading to concentration of risk to few annuity players
- Some of these players might themselves be state owned and subject to government guarantees
- The concentration risk in large providers can pose systematic risk to the financial system
- The government/regulator may be receiving lot of mis-selling complaints against the annuity providers
- For example selling normal annuities to pensioners who may have been eligible for impaired annuities with higher annuity amount
- On account of lobbying from financial groups/corporates who will benefit from such a change in regulation as funds would now be diverted towards them

[0.5 Marks for each] **[5 Max]****Solution 5:**

The change in the legislation would be expected to reduce the annuity take-up rate [0.5]

Customers normally like to have greater control over their assets rather than being tied up with a life time guaranteed annuity [0.5]

The level of the reduction in the take-up rate will depend upon a number of factors:

- The level of the fund value [0.5]
 - The lower the level of the fund value the more inclined policyholder's maybe to take the remaining amount as cash. [0.5]
 - The amount of fund may be low that the annuity product might not be available in the market [0.5]
 - Or the amount of annuity post conversion could be so low that it does not make sense to buy the annuity product [0.5]
- Personal preferences of policyholders [0.5]
 - customers may want cash to repay the mortgage or loans [0.5]
 - To fund any expenditure like child education, holidays etc. [0.5]
 - It will give them greater flexibility [0.5]

- Health of the policyholder [0.5]
 - In poor health may want cash for treatment [0.5]
 - Unlikely to buy immediate annuity if expected to die soon [0.5]
 - Impaired annuity might not be available [0.5]
 - Or impaired annuity might not be available at reasonable terms [0.5]
- Level of taxes [0.5]
 - The likely tax payable on taking the lump-sum in excess of allowed tax free [0.5]
 - The tax treatment of the annuity itself [0.5]
- Financial awareness of the policyholder [0.5]
 - Not aware that taking the cash will lead to no income during the older ages [0.5]
 - Annuity will lead to shift in the longevity risk to annuity provider [0.5]
 - They may not be able to comprehend the options available to them [0.5]
 - May have not sought professional advice [0.5]
- Others
 - If the annuity rates offered are competitive then may purchase annuity [0.5]
 - Alternative products available in the market [0.5]
 - The opportunity to invest in other high yielding assets [0.5]
 - Example: RIETs [0.5]
 - Tax treatment of such alternative products [0.5]
 - The level of the financial strength of the providers offering annuity [0.5]
 - If the financial strength of the providers is low then less will opt for an annuity [0.5]
 - The level of regular income from other investments/sources [0.5]
 - The financial status including other assets owned by the policyholder [0.5]
 - The financial support available from their children [0.5]
 - Any mass campaigns run by the annuity providers regarding the need to have guaranteed income till death [0.5]

[10 Max]**Solution 6:****Risks**

- Higher claim as compared to that assumed under-pricing leading to lower profitability [0.5]
- Rejection of claims leading to reputational risk for the insurer [0.5]
- Fraudulent claims particularly claim are seasonal and could go up during the down turn in the economy [0.5]
- May not have capabilities to underwrite the product leading to anti-selection and early claims [0.5]
- May not have relevant data to adequately price the feature [0.5]
- Risk of repeated claims over duration of the contract [0.5]
- Loose policy wordings mean confusion regarding what constitutes as own occupation, therefore claim could continue for longer duration than expected. [0.5]

Mitigation

- Having very strict definition own occupation and tight language of policy terms and conditions [0.5]
- Some of the causes of disability may be excluded from the coverage: [1.5]
 - Taking part in hazardous past times and activities
 - Criminal acts
 - Drug Abuse

- War and civil commutation
 - Self-inflicted injury
 - Imposing additional limits – for example, benefit payments limited to a maximum of 24 months [0.5]
 - Having strict underwriting standards of financial underwriting in order to avoid anti-selection [0.5]
 - Customers do not apply for levels of disability income disproportionate to his/her salary [0.5]
 - Having stringent claim verification procedure to determine fraud and non-disclosure [0.5]
 - Having adequate reinsurance for the waiver of premium benefit to reduce risk and provide technical expertise [0.5]
 - Closely monitoring the experience so that appropriate actions could be taken proactively [0.5]
 - For example increasing the respective charges, change of the target market, withdrawing the product [0.5]
 - Pricing the product prudently however premium/charge could be restricted due to competition or regulation [0.5]
 - Increasing the provisioning if the claim experience is expected to be worse than assumed [0.5]
 - The insurer may decide to restrict the types of occupation/industry and permitted/geographic who are permitted to buy contract but will impact the sales of the product [0.5]
- [10 Max]**
